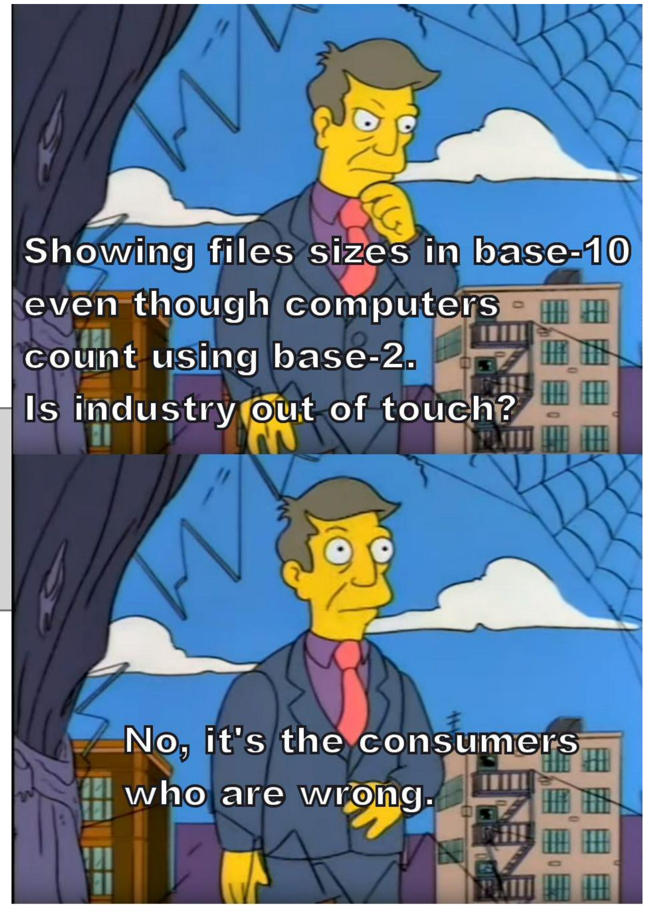
Digital Space

Help Lisa guide Homer to get his new HPE hPhone in the size he needs for all of his files and future storage needs.



Input

You will be given a list of items to store on a phone.

Each item will be listed in the format of NAME Q NN.nn
Where Q is the integer quantity of the item, and NN.nn are the average size of those file types in MiB.

STOP on a line by itself ends input.

```
Photos 713 5.83
Videos 65 222.04
Apps 2 512.15
WordDocs 10 1.23
Songs 321 6.72
STOP
```

Output

Truncate all output floating point numbers to 2 decimal places. Do not truncate until ready to output! Truncated numbers must be accurate to +/- 0.01.

Calculate how much space the user needs for their files in MiB, then translate that to MB and determine which phone model the user will need to store: their files + the OS files + have at least 500 MiB of free space available.

List the following in your output: the calculated space needed for files in MiB. The file size translated to MB. The size needed on the phone for files + 500MiB (translated to MB) + 200 MiB for OS files. The phone model chosen.

```
Files 21783.10 MiB
Files 22841.24 MB
Total 23575.24 MB
Need hPhone5 with 50 GB storage
```

Discussion

While the size of the files will be given in the actual units computers use (MiB, which is base-2), the size of the storage for phones in stores is almost always given in MB (which is base-10). The difference between those 2 start off small, but grow quickly apart as sizes grow. A quick size chart is given from digital file sizes in the discussion section. Your job is to determine what size phone (storage) the person needs to hold all of their files. Keep in mind that all phones take up at least 200MiB of the storage space for the operating system, and other phone-specific files. They should also have at least 500MiB of space free after all of the files have been loaded. E.G. if a phone has 1000MiB of space available, and the user has 800MiB of files to store, while technically the files would fit on that 1000MiB phone (200MiB for the OS files + the 800MiB for the person's files), there would be zero free space left, so the user would need to choose the next size up that would allow for at least 500MiB of free space. Available phone sizes are listed below. Your output should use the largest whole number units applicable for phone sizes. E.G. 1000MB should be output as 1GB.

MiB to MB size comparisons:

Unit	1s (bits)	byte (8 bits)	1000s	1,000,000s	1,000,000,000s
MiB	2^0	$2^0 * 8$	2^{10} = 1024	2^{20} = 1048576	2^{30} = 1073741824
MB	10^0	$10^0 * 8$	10^3 = 1000	10^6 = 1000000	10^9 = 1000000000

Digital Size Names:

Units	1000s	1,000,000s	1,000,000,000s	1,000,000,000s
Base-2	2^{10} = Kibibytes	2^{20} = Mibibytes	2^{30} = Gibibytes	2^{40} = Tebibytes
Base-10	10^3 = Kilobytes	10^6 = Megabytes	10^9 = Gigabytes	10^{12} = Terabytes

Phone Models Available:

Model	hPhone	hPhone1	hPhone2	hPhone3	hPhone4	hPhone5	hPhone6
Storage	100MB	250MB	500MB	1GB	10GB	50GB	75GB

Model	hPhone7	hPhone8	hPhone9	hPhoneX	hPhoneX1	hPhone12
Storage	100GB	150GB	200GB	500GB	1TB	2TB

This table is available in `phones.csv` in the files directory.

Additional Examples

Input 2	Output 2	Input 3	Output 3
Photos 9943 18.99 Videos 3155 109.72 Apps 82 123.45 OfficeFiles 55 6.93 Songs 15 5.28 Ringtones 6 1.22 Misc 85 55.65 STOP	Files 550304.98 MiB Files 577036.60 MB Total 577770.60 MB Need hPhoneX1 with 1 TB storage	Photos 3 22.12 Videos 0 111.11 Apps 1 39.99 Songs 0 8.13 STOP	Files 106.35 MiB Files 111.51 MB Total 845.51 MB Need hPhone3 with 1 GB storage